



Harasees Kaur

PhD Scholar (CSE)

Department of Computer Science and Engineering

Indian Institute of Technology Ropar

+91-6239315280

harasees.24csz0001@iitrpr.ac.in

SID: 2024CSZ0001

India

RESEARCH SUMMARY

I am a Ph.D. scholar at IIT Ropar working in the area of High-Performance Computing and Distributed Systems with a focus on scalability-aware and communication-centric design and evaluation of MPI collective communication. My research investigates communication bottlenecks across small-scale educational clusters and large-scale HPC systems, with particular emphasis on performance modeling, trace-driven simulation, and optimization for distributed AI workloads.

EDUCATION

Degree	Institute/Board	CGPA / %	Year
Ph.D. (CSE)	Indian Institute of Technology Ropar	9.23 (Coursework)	2024–Present
M.Tech (CSE)	Punjab Engineering College, Chandigarh	8.64	2022–2024
B.Tech (CSE)	Chitkara University, Rajpura	9.74	2018–2022
Class XII	CBSE, CAKFMSS School	81%	2018
Class X	CBSE, CAKFMSS School	10.0 CGPA	2016

PHD SUPERVISION

- **Primary Supervisor:** Dr. Jagpreet Singh, Assistant Professor, CSE, IIT Ropar
- **Co-Supervisor:** Dr. Balwinder Sodhi, Associate Professor, CSE, IIT Ropar

RESEARCH INTERESTS

High-Performance Computing and Distributed Systems; MPI Collective Communication and Synchronization; Communication Performance Analysis and Trace-Driven Modeling; Scalable Parallel Algorithms; Distributed AI and Data-Parallel Training; Topology-Aware and Multicast Communication; Performance Modeling for Heterogeneous Systems; HPC Education

PUBLICATIONS

- Harasees Kaur, Sankalp Shashi, Umesh Konduru, Sahil Kumar, Balwinder Sodhi, BS Sanjeev, Jagpreet Singh. 2025. “**Analyzing Communication Performance From Testbeds to HPC Systems: A Lab Course**”, Proceedings of 7th Workshop on Education for High Performance Computing (EduHiPC 2025) in conjunction with HiPC, Hyderabad, India, December 2025.
- Harasees Kaur, Padmavati Khandnor, Kanu Goel, **Machine Learning Based Hybrid Approach for Schizophrenia Diagnosis Using Non-Stationary EEG Signals**, International Journal of Biomedical Engineering and Technology (IJBET), 2025. *Published*.
- Harasees Kaur, Padmavati Khandnor, Kanu Goel, **A Novel Approach to EEG-Based Schizophrenia Detection Using VMD and Machine Learning Algorithms**, 5th International Conference on Data Science and Applications (ICDSA 2024), MNIT Jaipur, India. *Published*.

TEACHING EXPERIENCE

Teaching Assistant – HPC Lab Course (CS565)

Department of Computer Science and Engineering

Jan 2026 – Present

IIT Ropar

- Supporting the design and delivery of the High-Performance Computing Lab course.
- Assisting students with MPI programming, collective communication, and parallel experiments.
- Guiding hands-on sessions on cluster usage, performance evaluation, and scalability studies.

Teaching Assistant – Operating Systems
Department of Computer Science and Engineering

July 2025 – Nov 2025
IIT Ropar

- Conducted lab sessions and tutorials for Operating Systems.
- Assisted students with synchronization, scheduling, and memory management concepts.

Teaching Assistant – Programming Paradigms and Pragmatics
Department of Computer Science and Engineering

Jan 2025 – May 2025
IIT Ropar

- Conducted practice sessions and tutorials for undergraduate students.
- Assisted in explaining programming paradigms, problem-solving techniques, and coding best practices.

Teaching Assistant – Operating Systems
Department of Computer Science and Engineering

July 2024 – Nov 2024
IIT Ropar

- Conducted lab sessions and tutorials for Operating Systems.
- Supported evaluation and doubt-solving sessions for coursework and labs.

PROJECTS

HPC in AI/ML (Ongoing)
Ph.D. Research, IIT Ropar

2024 – Present

- Working on efficient collective communication to improve scalability of distributed AI/ML workloads.
- Focus areas include communication overhead reduction and MPI collective performance characterization.

Machine Learning-Based Schizophrenia Diagnosis Using EEG Signals
M.Tech Thesis Project, PEC Chandigarh

2023 – 2024

- Developed ML pipelines for non-stationary EEG analysis using VMD and feature engineering.
- Designed hybrid classification approach to improve diagnosis performance.

VOLUNTEERING AND OUTREACH

- Volunteer, **SYNTACS Conference**, IIT Ropar.
- Volunteer, **Conference on Computer Vision and Image Processing (CVIP)**, IIT Ropar.
- Anchor and Volunteer, **Swasth Nari Shakti Parivar Campaign** (Government of India initiative), organized at IIT Ropar.
- Volunteer, **Faculty Development Program on Deep Learning: Fundamentals and Case Studies in Healthcare**, Punjab Engineering College, Chandigarh.

INDUSTRY EXPERIENCE

Software Engineer – Java Developer
OOPS Info Solutions Pvt. Ltd.

2022 – 2023
Chandigarh, India

- Implemented and customized an ERP system to streamline business processes and improve data consistency.
- Worked on feature development, debugging, and performance improvements for production modules.

TECHNICAL SKILLS

- **Programming:** C, C++, Python, Java
- **HPC / Parallel Computing:** MPI, OpenMP, Collective Communication
- **Systems and Tools:** Linux, Slurm, OpenMPI, OSU Micro-Benchmarks, Netgauge

KEY COURSES

- **Core CS:** Algorithms & Data Structures, Operating Systems, DBMS, Computer Networks
- **HPC / Systems:** Parallel Computing, Distributed Systems